



COVENANT UNIVERSITY

CANAANLAND, KM 10, IDIROKO ROAD
P.M.B 1023, OTA, OGUN STATE, NIGERIA

TITLE OF EXAMINATION: B.SC. EXAMINATION
COLLEGE: College of Science and Technology

DEPARTMENT: Computer and Information Sciences

SESSION: 2019/2020

COURSE CODE: CSC213

COURSE TITLE: Structured Programming

INSTRUCTION: Answer any **FOUR (4)** Questions

SEMESTER: ALPHA

CREDIT UNIT: 3

TIME: 3 hours

QUESTION 1

- a. With the aid of examples, mention the **Five (5)** relational operators in Python [2.5Marks]
- b. Write an if-else statement that determines whether a variable called **MyValues** that accepts input from the user is outside the range of **11** to **57**. If the variable's value is outside this range it should display "*Invalid Points*"; Otherwise, it should display "*Valid Points*." [2 Marks]
- c. What will the output of the following code display? [1 Mark]

```
values = ['A', 14, 601, '88', 10]
print (values[1:3])
```

- d. Identify the Error(s) in the Following Code Segment [3 Marks]

```
def main():
    prod_nums = ['V475', 'F987', 'Q143', 'R688']
    search = input('Enter a product number: ')
    if sarch in prod_nums:
        print(search, 'was found in the list.')
    else:
        print(search, 'was not found in the list.')
    main()
```

- e. Design a program that uses a loop to build a list named **valid_numbers** that contains only the numbers between 0 and 100 from the numbers list below. The program should then determine and display the **maximum**, **total** and **average** of the values in the **valid_numbers** list. [9 Marks]

Numbers = [45, 13, 290, 105, -20, 2, 67, 17, -12, 51, 80]

*check = every number in
integers[]
for n in range (check + 1)*

max = 0

*for n in numbers
if n < range (100)
max = n*

size = len (valid - n)

*for i in range (size)
valid_numbers = n*

avg = total / len

Question 2

- a. What is a Boolean function in Python? [1 Mark]
- b. Write a statement that creates a dictionary containing the following key-value pairs: [2 Marks]

'CST'	:1
'CBSS'	:2
'CLDS'	:3
'COE'	:4

- c. Write a loop that calculates the **Total** of the following series of numbers: [3.5 Marks]

$$\frac{1}{50} + \frac{2}{49} + \frac{3}{48} + \dots + \frac{50}{1}$$

- d. Spot the Error(s) in the following Code Segment [3 Marks]

```
def main():
    outfile = openfile('citiesInNigeria.txt', 'r')
    outfile.write(Abuja\n)
    outfile.write(Paris\n)
    outfile.write(Kaduna\n)
    outfile.closefile()
    main()
```

- e. Assume a file named **HarvestNumbers.txt** containing an unspecified series of integer values exists on the computer's disk. Write a program that reads the numbers from the file and calculates the average of all the numbers. [8 Marks]

Question 3

- a. Differentiate between a **Condition-Controlled** and **Counter-Controlled** loop? Mention and Show examples of each loop type. [2.5 Marks]
- b. Write an **if** statement that adds **500** to a variable **first_1**, and adds **400** to the variable **second_2** if the values entered as input into variable **inputVar** is greater than **100**, otherwise, it deducts **300**, from **first_1** and **150** from **second_2**. [2 Marks]
- c. Identify the Error(s) in the following lines of Code [3 Marks]

```
def main():
    name_list = []
    again == 'y'
    while again = 'y':
        name = input('Enter a name: ')
        Append the name to the list=list().name
        name_list.append(name)
        print('Do you want to add another name?')
        again = input('y = yes, anything else = no: ')
        print()
    print('Here are the names you entered.')
    for name in name_list:
        print(name)
    main()
```

from [0:0]
line [0]

first_1 = 500 + 400 = 900
second_2 = 400 + 150 = 550

Write a program that gets strings containing a person's first and last name as separate values, and then displays their "initials", "name in address book", "username", and "reversed form". For example, if the user enters a first name of "David" and a last name of "Oyedepo", the program should display "D.O", "David OYEDEPO", "davidoyedepo", and "OPEDEYO Divad". [10 Marks]

Question 4

a. With the aid of a sample code, describe and infinite loop? [1.5Marks]

b. What will the following code display? [1 Mark]

```
MyDict = {1: [1, 2, [3, 4], 5, 6]}
print (MyDict [0][2][1])
```

c. Given the following statement:

```
levels = 'Beginner, Average, Advanced, Expert'
```

Write a statement that splits this string, creating the following list: ['Beginner', 'Average', 'Advanced', 'Expert'] [1 Marks]

d. Identify the Error(s) in the following lines of code [3 Marks]

```
def main():
    print('I have a message for you.').
    message():/
    print('Goodbye!')

message():
    print('I am a Student of Greatness',)
    print('A Royalty in Hebron'.),
main()
```

e. Write out the output of the correct version of the code in above [2 Marks]

f. The date **June 10, 1960**, is special because when it is written in the following format, the month times the day equals the year: **6 / 10 / 60**. Design a program that asks the user to enter a month (in shortened form, e.g. Oct), a day in numeric form (e.g. 07), and a four-digit year (e.g. 1978). The program should then determine whether the month times the day equals the year. If so, it should display a message saying the date is **Special**. Otherwise, it should display a message saying the date is **Not Special**. (Hint: use Dictionary and Strings). [9 Marks]

Question 5

a. What will the following program display? [1.5Marks]

```
month =
month * 10 = month [month]
```

```
def main():
    x = 1
    y = 3.4
    print(x, y)
    change_us(x, y)
    print(x, y)
def change_us(a, b):
    a = 0
    b = 0
    print(a, b)
main()
```

Question
a. Write code that repeats a program.
b. What will

- b. Write a program that opens an output file with the filename **bibleinfo.txt**, writes the name of a bible book, a bible character, and a bible city to the file on separate lines, then closes the file. [2 Marks]

- c. Show the output of the following code

```
for i in range(2,10):
    values = [pow(i,2)] * (i-1)
    print(values)
```

[4 Marks]

Handwritten calculations for the output of the code in part c:

$2^2 \times 1 = 4$	$7^2 \times 6 = 294$
$3^2 \times 2 = 18$	$8^2 \times 7 = 504$
$4^2 \times 3 = 48$	$9^2 \times 8 = 648$
$5^2 \times 4 = 100$	
$6^2 \times 5 = 180$	

- d. Identify the Error(s) in the Code Below [3 Marks]

```
num_students = int(input('How many students do you have?'))
num_test_scores = int(input('How many test scores per student?'))
for students in range(num_students):
    total = 0.0
    print('Student number', student + 1)
    print('-----')
    for test_num in range(num_test_scores):
        print('Test number', test_num + 1, end='')
        score = float(input(': '))
        total += score
        average = total / num_test_scores
        print('Average for student #', student + 1, 'is:', average)
print()
```

- e. When an object is falling because of gravity, the following formula can be used to determine the distance the object falls in a specific time period:

$$d = \frac{1}{2}gt^2$$

The variables in the formula are as follows: **d** is the distance in meters, **g** is 9.8, and **t** is the amount of time, in **seconds**, that the object has been falling. Write a function named **falling_distance** that accepts an object's falling time (in **minutes**) as an argument. The function should return the distance, in meters, that the object has fallen during that time interval. Write a program that calls the function in a loop that passes the values 1 through 10 as arguments and displays the return value. [7 Marks]

Handwritten notes for part e:

$g = 9.8$
 $d = \frac{1}{2}gt^2$
 time = minutes
 fall dist (time)

Handwritten code for part e:

```
def falling_distance(t):
    d = 0.5 * g * t**2
    return d
```

Handwritten code for part e:

```
for i in range(1, 11):
    time = i
    time = time / 60
```

Question 6

- a. Write code that contains a loop that repeats 5 times nested within a loop that repeats 4 times, and state the total iterations that would occur in such a program. [1.5 Marks]

- b. What will be the output when the following code is executed? [1 Mark]

```
months = {'Jan':1, 'Feb':2, 'Mar':3}
print (months[2])
```

- c. Look at the following statement: **NewLaw** = ['L', 'O', 'V', 'E'], What values are stored in NewLaw [-1], NewLaw [-3], and NewLaw [-2] [3 Marks]

- d. Identify the Error(s) in the following Code [3 Marks]

```
def main():
    first_age = int(input('Enter your age: '))
    second_age = int(input("Enter your friend's age: "))
    total = sum(first_age, second_age);
    print('Together you are', total, 'years old'.)
sum(num1, num2):
    result = first_age + second_age
    return
main()
```

- e. Write a GUI program that converts Celsius temperatures to Fahrenheit temperatures. The user should be able to enter a Celsius temperature, click a button, then see the equivalent Fahrenheit temperature. Use the following formula to make the conversion: **F** is the Fahrenheit temperature, and **C** is the Celsius temperature. [9 Marks]

$$F = \frac{9}{5}C + 32$$